Wilfried Tapsoba

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SUMMARY

B.S. Computer Engineering student at Milwaukee School of Engineering with a concentration on electrical engineering, hardware validation, and embedded systems. Experienced in troubleshooting and diagnosing hardware issues at the board level through internship and project work. Proficient in using electrical test lab equipment for schematic design and PCB layout. Self-motivated individual with a published research background in AI systems and a commitment to solving open-ended hardware challenges. Skills: 3D Printing, MATLAB, AutoCAD, Java, Python, C, Quartus, VHDL, ARM Assembly, Embedded Systems, Arduino, Circuit Design, Research, Data Analysis, Project Management.

EDUCATION

B. S. Computer Engineering | Milwaukee School of Engineering, Milwaukee, WI | GPA: 3.5 | May 2026
AI for Emerging Applications Certificate | Milwaukee School of Engineering | December 2025
Associate of Science | Madison College, Madison, WI | GPA: 3.84 | May 2023

SKILLS

Hardware Test and Validation: Oscilloscopes, Logic Analyzers, DMM, Power Supplies, Signal Generator.
Programming and Development: Python, C, Java, MATLAB, ARM Assembly, VHDL, Software Development, Data Structures, PLC Programming, AutoCAD
Operating System: Linux, Computer Architecture, Bash, Git.
Hardware and Systems Design: Intel Quartus, Raspberry Pi, Arduino Devices, STM32CubeIDE, Electrical Circuits, Multisim
Leadership: Synthesizes complex information for succinct team and leadership briefings; fosters team cohesion by aligning efforts toward common goals; effectively communicates vision, sets achievable objectives, and delegates responsibilities to drive success.

INTERNSHIP EXPERIENCE

Controls Engineer Co-Op | Husco | January - August 2025 | Full-time.

- Performed microcontroller feature validation, tested analog, digital, and multi-function inputs to ensure hardware-software reliability.
- Conducted root cause analysis on hardware through thermal paste effectiveness tests and internal motor temperature assessments.
- Built and verified prototypes for new products, evaluating the operational theory of circuits to ensure design conformance.
- Developed MATLAB code for hardware-interfacing software and authored comprehensive technical documentation to streamline adoption.

Innovation and Entrepreneurship Virtual Internship | Excelerate | January - March 2025

- Conducted market research to identify gaps and opportunities for business solutions.
- Performed feasibility analysis, including revenue projections and cost assessments.
- Gained hands-on experience in business strategy development and innovation consulting.

PROJECT EXPERIENCE

Embedded Project: Smart Door Control System | Milwaukee School of Engineering | Team of 4

- Developed an embedded security system using STM32, PIR motion sensors, and audible alarms to detect and respond to unauthorized access.
- Built real-time touchscreen GUI with TouchGFX, C++ (RTOS) for system control, alerts, configurable sensor timing.
- Prototyped RFID authentication and debugged hardware–software integration using logic analysis tools.

Embedded Fabrication Project: Wi-Fi/Bluetooth Alarm Clock
Goal: Design and create a PCB for an alarm clock with speakers and wireless capabilities

- Create a 2-layer PCB to run the electronics for the alarm clock, along with the inputs and outputs for external components.
- Integrated an ESP32-WROOM module to allow the alarm clock to update the time using the internet and receive weather data.
- Implemented I^2C to communicate with a real-time clock for offline timekeeping and a GPIO integrated circuit.
- Utilized a HUB75 LED matrix display for the user interface.

Digital Logic Project: Autonomous Robot Management

- Managed an autonomous robot through a digital circuit, completing motion control truth tables and a 7-segment display decoder.
- Assembled the controller and decoders using structural VHDL.
- Tested the circuit using simulation waveforms to verify behavior and modified it for optimal performance.

Artificial Intelligence Research Project: Multi-Agent Problem Solving with Large Language Models

- Developed a multi-agent framework that allows LLMs to simulate thought and action interpretation
- Modeled and analyzed how groups of LLM-powered agents collaborate to solve problems
- Evaluated the effectiveness of the multi-agent systems in both simulated environments, demonstrating the potential for LLMs to mimic human-like collaborative problem-solving. Published paper: https://arxiv.org/abs/2409.13753

LEADERSHIP EXPERIENCE | CO-CURRICULAR INVOLVEMENT | COMMUNITY SERVICE

Artificial Intelligence Research Team member | AI-Club, MSOE | September 2023 - Present
National Society of Black Engineer | MSOE chapter | September 2023 - Present
STEM Ambassador | Madison College | September 2021 - August 2023

- Improve the organization and accessibility of internal documentation through effective formatting and categorization; Perform an inventory of equipment, and organize all information in Excel.
- Develop training documents and materials for new ambassadors.

Student Government President | Madison College, Madison, WI | May 2021 – May 2023

- Attended the National Legislative Summit February 5-8 in DC, met with US. Rep. Mark Pocan and Senator Tammy Baldwin, and advocated for educational initiatives that will improve students' experience.
- Attended and voted on business at the Academic Council and the Student Affairs Council meetings, provided feedback and insights on behalf of all students at Madison College.

STEM Club Coordinator | Madison College, Madison, WI | September 2021 – May 2023
Volunteering | Catholic Multicultural Center, English tutor| STEM tutor, Madison, WI, Fall 2022 | over 100hrs.

WORK EXPERIENCE

STEM Lab Instructor | Milwaukee School of Engineering | September 2023- Present | 12 hrs per wk

- Support evening programs as an instructor and mentor for coding clubs and STEM initiatives.
- Lead STEM projects in electronics, circuit design, 3D modeling, programming, and engineering.
- Plan and teach STEM-related projects to youth in Milwaukee; Support youth with an interest in STEM.

STEM Instructional Support Technician | Madison College | May 2023 - August 2023

- Assisted in the design and development of items on SOLIDWORKS to promote access in STEM.
- Repaired and maintained equipment, did inventory, and brainstormed project ideas for the center.
- Revised the website, provided suggestions with a new one I programmed using JavaScript and CSS.